

Quiz Performance Report

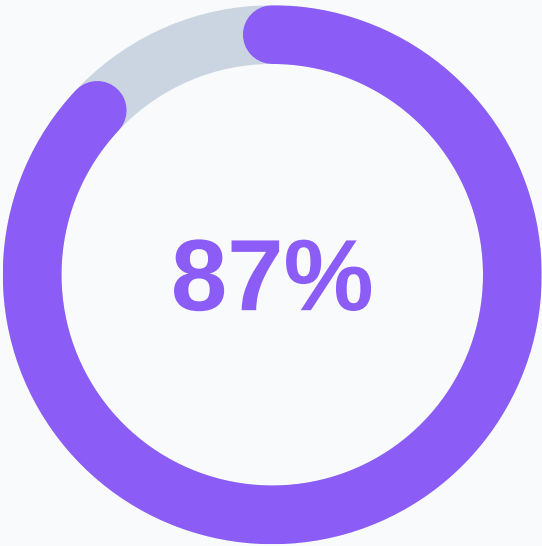
Performance Summary

A detailed breakdown of your quiz results and performance

Total Questions	28
Questions Attempted	28
Questions Passed	25
Questions Failed	3

Overall Score

Your overall quiz score percentage



Quiz Questions

1. Which method is used to connect an Express application to a MongoDB database? **MEDIUM**

`express.connect()`

`mongoose.connect()`

Correct Answer

`mongodb.attach()`

`mongoose.attach()`

Explanation

Mongoose.connect establishes a connection to the MongoDB database.

2. In Express, how can you define a route that handles GET requests to '/api/users'? **EASY**

`app.router('/api/users')`

`app.serve('/api/users')`

`app.get('/api/users', function(req, res){})`

Correct Answer

`app.fetch('/api/users', function(req, res){})`

Explanation

The app.get method in Express is used to define a route that handles GET requests.

3. Which statement correctly initializes a new MongoDB collection using Mongoose? MEDIUM

`new Mongo.Collection('collectionName')`

`db.collection('collectionName')`

`const Collection = mongoose.model('CollectionName', new mongoose.Schema({}))`

Correct Answer

`create.collection('collectionName')`

Explanation

The `mongoose.Schema` method is used to define a new collection model in MongoDB.

4. What is the role of Babel in frontend development? EASY

Babel is used to create layouts using templates.

Babel is a JavaScript runtime environment.

Babel transpiles ES6+ code to JavaScript ES5 compatible code.

Correct Answer

Babel is a CSS preprocessor like Sass.

Explanation

Babel transpiles ES6 code to JavaScript compatible across all browsers.

5. Which JavaScript ES6 feature allows destructuring of arrays or objects?

EASY

Arrow functions

Promises

Destructuring

Correct Answer

Template literals

Explanation

Destructuring allows unpacking of arrays or properties from objects using an efficient syntax.

6. What is the primary purpose of the Node.js runtime environment?

EASY

To compile React components.

To run Java applets in browsers.

To execute JavaScript code outside of a browser.

Correct Answer

To create animated graphics using WebGL.

Explanation

Node.js is primarily used for non-blocking, event-driven servers because of its asynchronous nature.

7. Which cloud service is popular for deploying Node.js applications?

EASY

Amazon S3

Firebase

Your Answer

AWS Elastic Beanstalk

Correct Answer

Google Analytics

Explanation

AWS Elastic Beanstalk is a popular service for deploying Docker and Node.js applications.

8. What would be a result of invoking the following JavaScript code snippet:

EASY

`[1, 2, 3].map(x => x * 2)?`

[1, 2, 3]

[2, 4, 6]

Correct Answer

undefined

[1, 4, 9]

Explanation

The `map()` function applies a function to each element and returns a new array.

9. What is the two-way binding often used in React for form elements?

MEDIUM

Using JQL syntax

Synchronizing the input with the state using handlers

Correct Answer

Using Express to handle bindings

Binding input values directly to state without handlers

Explanation

Two-way binding in React is achieved by tying the input element to the component's state, and vice versa.

10. How do you define an arrow function in JavaScript ES6?

EASY

function() => { return x; }

const func = () => x

Correct Answer

arrow func() return x

function = () => { x; }

Explanation

The arrow function syntax is more concise compared to regular function expressions.

11. In MongoDB, which operation would you use to retrieve a specific document from a collection?

EASY

insertOne

updateOne

find

Correct Answer

delete

Explanation

The find method retrieves documents matching a query in MongoDB.

12. What happens if you don't provide a dependency array in useEffect?

EASY

Runs once

Runs only on unmount

Runs after every render

Correct Answer

Throws error

Explanation

Without dependencies, useEffect runs after every render, causing performance issues.

13. What is the problem with this MongoDB query?

EASY

```
User.find({ age: { $gt: "25" } })
```

Syntax error

String vs number mismatch

Correct Answer

Too fast

Always returns empty

Explanation

Comparing string "25" instead of number ' incorrect results.

14. What is the issue with this API call in React?

MEDIUM

```
useEffect(() => {  
  fetch('/api/data').then(res => res.json()).then(setData);  
}, []);
```

No issue

Runs multiple times

Syntax error

Your Answer

Memory leak if component unmounts before response

Correct Answer

Explanation

If component unmounts ' state update causes warning ' use cleanup/abort controller.

15. What does cluster module do in Node.js?

HARD

Handles DB

multi-core scaling

Correct Answer

Improves frontend

Adds caching

Explanation

Enables multi-core scaling by spawning worker processes.

16. What is the issue with this Express route?

MEDIUM

```
app.get('/user', (req, res) => {  
  res.send("Hello");  
  res.send("Again");  
});
```

Both responses are buffered and merged into a single HTTP response before being sent

The second `res.send()` triggers an `ERR_HTTP_HEADERS_SENT` error because headers were already finalized after the first response

Correct Answer

Express silently ignores subsequent `res.send()` calls after the first successful response

The second `res.send()` overwrites the first response due to asynchronous execution of the event loop

Explanation

Once `res.send()` is called, headers + body are sent and the response is closed. Any further attempt to send another response results in "Cannot set headers after they are sent" (`ERR_HTTP_HEADERS_SENT`).

17. What is the output behavior in this React code?

MEDIUM

```
const [count, setCount] = useState(0);  
setCount(count + 1);  
setCount(count + 1);
```

count becomes 2

count becomes 1

Correct Answer

count becomes 0

Throws error

Explanation

Both updates use same stale value ' React batches ' final = 1.

18. What is the result of this MongoDB query?

EASY

```
db.users.find({ age: { $exists: true } })
```

Returns users with age > 0

Returns users where age is not null

Returns users where age field exists

Correct Answer

Returns all users

Explanation

\$exists checks field presence, not value.

19. What happens in this Node.js code?

MEDIUM

```
setTimeout(() => console.log("A"), 0);  
console.log("B");
```

A then B

B then A

Correct Answer

Only A

Random

Explanation

setTimeout runs after call stack ' async.

20. What does this Express route return?

HARD

```
app.get('/test', (req, res) => {  
  return res.json({ a: 1 });  
  res.json({ b: 2 });  
});
```

{a:1, b:2}

{b:2}

{a:1}

Correct Answer

Error

Your Answer

Explanation

return exits function ' second line never runs.

21. What is the issue with this key usage?

EASY

```
items.map(item => <div key={Math.random()}>{item}</div>)
```

No issue

Causes re-render issues

Correct Answer

Improves performance

Syntax error

Explanation

Keys must be stable, random breaks diffing.

22. What type of database is MongoDB?

EASY

Relational

Document-based NoSQL

Correct Answer

Graph database

Key-value only

Explanation

MongoDB stores data as JSON-like documents (BSON).

23. What does findOne() return?

EASY

All documents

First matching document

Correct Answer

Array of documents

Error

Explanation

Returns only one document.

24. What is Node.js built on top of?

EASY

SpiderMonkey engine

V8 JavaScript engine

Correct Answer

Chakra engine

JVM

Explanation

Node.js uses Google's V8 engine (also used in Chrome) to execute JavaScript outside the browser at high speed.

25. Which module is used to read and write files in Node.js?

EASY

http

fs

Correct Answer

stream

buffer

Explanation

The fs (File System) module lets you interact with the file system — reading, writing, deleting, and watching files.

26. What is the main purpose of the useState hook in React?

EASY

To subscribe to state changes in the Redux store.

To handle HTTP requests in React components.

To define the initial state and update logic in a functional component.

Correct Answer

To style components using CSS-in-JS.

Explanation

The useState hook allows functional components to manage local state.

27. In React, how can you pass data from a parent component to a child component? EASY

Using state

Using props

Correct Answer

Using context

Using ref

Explanation

Props are used to pass data from a parent component to its child component in React.

28. Which CSS property is used to create a flex container? EASY

flex-direction: row

display: grid

display: flex

Correct Answer

flex-wrap: wrap

Explanation

display: flex; is the property used to define a flex container.